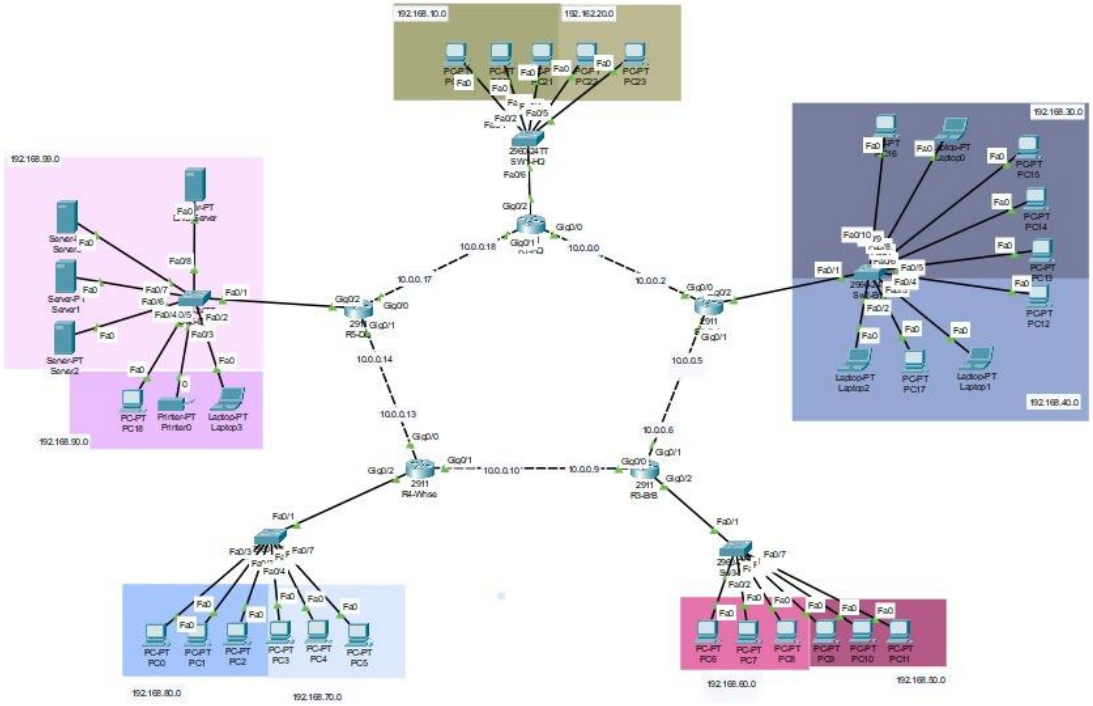
Final project



R1-HQ :

- Basic Configurations

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1-HQ
R1-HQ(config)#banner motd ""AUTHORIZED ACCESS ONLY - GLN
R1-HQ(config)#banner motd "AUTHORIZED ACCESS ONLY - GLN NETWORK"
R1-HQ(config)#enable secret class
R1-HQ(config)#service password-encryption
R1-HQ(config)#line con 0
R1-HQ(config-line)#pass cisco
R1-HQ(config-line)#login
R1-HQ(config-line)#ex
% Ambiguous command: "ex"
R1-HQ(config-line)#exit
R1-HQ(config)#line vty 0 4
R1-HQ(config-line)#pass cisco
R1-HQ(config-line)#login
R1-HQ(config-line)#exit
R1-HQ(config)#ex
R1-HQ#
%SYS-5-CONFIG_I: Configured from console by console

R1-HQ#clock set 9:00:00 28 sep 2026
R1-HQ#
```

- ROAS and WAN interfaces

```
R2-BrA>en
Password:
R2-BrA#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2-BrA(config)#int g0/2
R2-BrA(config-if)#no shut

R2-BrA(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

R2-BrA(config-if)#ex
R2-BrA(config)#int g0/2.30
R2-BrA(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.30, changed state to up

R2-BrA(config-subif)#encapsulation dot1q 30
R2-BrA(config-subif)#ip add 192.168.30.1 255.255.255.0
R2-BrA(config-subif)#ex
R2-BrA(config)#int g0/2.40
R2-BrA(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2.40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.40, changed state to up

R2-BrA(config-subif)#encapsulation dot1q 40
R2-BrA(config-subif)#ip add 192.168.40.1 255.255.255.0
R2-BrA(config-subif)#ex
R2-BrA(config)#int g0/0
R2-BrA(config-if)#no shut

R2-BrA(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2-BrA(config-if)#ip add 10.0.0.2 255.255.255.252
R2-BrA(config-if)#ex
R2-BrA(config)#int g0/1
R2-BrA(config-if)#ip add 10.0.0.5 255.255.255.252
R2-BrA(config-if)#no shut

R2-BrA(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

R2-BrA(config-if)#ex
```

Copy Pas

- EIGRP Configurations

```
R1-HQ>en
Password:
R1-HQ#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1-HQ(config)#router eigrp 100
R1-HQ(config-router)#network 192.168.10.0
R1-HQ(config-router)#network 192.168.20.0
R1-HQ(config-router)#network 10.0.0.0
R1-HQ(config-router)#network 10.0.0.16
R1-HQ(config-router)#no auto-summary
R1-HQ(config-router)#ex
R1-HQ(config)#
```

- DHCP and DNS Servers

```
R1-HQ(config)#ip dhcp excluded-address 192.168.10.0 192.168.10.10
R1-HQ(config)#ip dhcp excluded-address 192.168.20.0 192.168.20.10
R1-HQ(config)#ip dhcp pool pool-Admin
R1-HQ(dhcp-config)#network 192.168.10.0 255.255.255.0
R1-HQ(dhcp-config)#default-router 192.168.10.1
R1-HQ(dhcp-config)#dns-server 192.168.90.10
R1-HQ(dhcp-config)#ex
R1-HQ(config)#ip dhcp pool pool-Finance
R1-HQ(dhcp-config)#network 192.168.20.0 255.255.255.0
R1-HQ(dhcp-config)#default-router 192.168.20.1
R1-HQ(dhcp-config)#dns-server 192.168.90.10
R1-HQ(dhcp-config)#ex
R1-HQ(config)#
```

- Show IP Route

```
R1-HQ>en
Password:
R1-HQ#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
D       10.0.0.4/30 [90/3072] via 10.0.0.2, 00:13:32, GigabitEthernet0/0
D       10.0.0.8/30 [90/3328] via 10.0.0.17, 00:13:32, GigabitEthernet0/1
                    [90/3328] via 10.0.0.2, 00:13:32, GigabitEthernet0/0
D       10.0.0.12/30 [90/3072] via 10.0.0.17, 00:13:32, GigabitEthernet0/1
C       10.0.0.16/30 is directly connected, GigabitEthernet0/1
L       10.0.0.18/32 is directly connected, GigabitEthernet0/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/2.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/2.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/2.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/2.20
D    192.168.30.0/24 [90/28416] via 10.0.0.2, 00:13:32, GigabitEthernet0/0
D    192.168.40.0/24 [90/28416] via 10.0.0.2, 00:13:32, GigabitEthernet0/0
D    192.168.50.0/24 [90/28672] via 10.0.0.2, 00:13:32, GigabitEthernet0/0
D    192.168.60.0/24 [90/28672] via 10.0.0.2, 00:13:32, GigabitEthernet0/0
D    192.168.70.0/24 [90/28672] via 10.0.0.17, 00:13:32, GigabitEthernet0/1
D    192.168.80.0/24 [90/28672] via 10.0.0.17, 00:13:32, GigabitEthernet0/1
D    192.168.90.0/24 [90/28416] via 10.0.0.17, 00:13:32, GigabitEthernet0/1
D    192.168.99.0/24 [90/28416] via 10.0.0.17, 00:13:32, GigabitEthernet0/1

R1-HQ#
```

R2-BrA :

- Basic Configurations

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2-BrA
R2-BrA(config)#banner motd "AUTHORIZED ACCESS ONLY - GLN NETWORK"
R2-BrA(config)#enable secret class
R2-BrA(config)#service password-encryption
R2-BrA(config)#line con 0
R2-BrA(config-line)#pass cisco
R2-BrA(config-line)#hogin
                   ^
% Invalid input detected at '^' marker.

R2-BrA(config-line)#login
R2-BrA(config-line)#ex
% Ambiguous command: "ex"
R2-BrA(config-line)#exit
R2-BrA(config)#line vty 0 4
R2-BrA(config-line)#pass cisco
R2-BrA(config-line)#login
R2-BrA(config-line)#exit
R2-BrA(config)#ex
R2-BrA#
%SYS-5-CONFIG_I: Configured from console by console

R2-BrA#clock set 9:00:00 28 sep 2026
R2-BrA#
```

-ROAS and WAN interfaces

```
R2-BrA>en
Password:
R2-BrA#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2-BrA(config)#int g0/2
R2-BrA(config-if)#no shut

R2-BrA(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

R2-BrA(config-if)#ex
R2-BrA(config)#int g0/2.30
R2-BrA(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.30, changed state to up

R2-BrA(config-subif)#encapsulation dot1q 30
R2-BrA(config-subif)#ip add 192.168.30.1 255.255.255.0
R2-BrA(config-subif)#ex
R2-BrA(config)#int g0/2.40
R2-BrA(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2.40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.40, changed state to up

R2-BrA(config-subif)#encapsulation dot1q 40
R2-BrA(config-subif)#ip add 192.168.40.1 255.255.255.0
R2-BrA(config-subif)#ex
R2-BrA(config)#int g0/0
R2-BrA(config-if)#no shut

R2-BrA(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2-BrA(config-if)#ip add 10.0.0.2 255.255.255.252
R2-BrA(config-if)#ex
R2-BrA(config)#int g0/1
R2-BrA(config-if)#ip add 10.0.0.5 255.255.255.252
R2-BrA(config-if)#no shut

R2-BrA(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

R2-BrA(config-if)#ex
```

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- EIGRP Configurations

```
R2-BrA>
R2-BrA>en
Password:
R2-BrA#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2-BrA(config)#router eigrp 100
R2-BrA(config-router)#no auto
R2-BrA(config-router)#network 192.168.30.0
R2-BrA(config-router)#network 192.168.40.0
R2-BrA(config-router)#network 10.0.0.0
R2-BrA(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.0.1 (GigabitEthernet0/0) is up: new adjacency

R2-BrA(config-router)#network 10.0.0.4
R2-BrA(config-router)#ex
R2-BrA(config)#
```

- DHCP and DNS Servers

```
R2-BrA(config)#
R2-BrA(config)#ip dhcp excluded-address 192.168.30.0 192.168.30.10
R2-BrA(config)#ip dhcp excluded-address 192.168.40.0 192.168.40.10
R2-BrA(config)#ip dhcp pool pool-Sales
R2-BrA(dhcp-config)#network 192.168.30.0 255.255.255.0
R2-BrA(dhcp-config)#default-router 192.168.30.1
R2-BrA(dhcp-config)#dns-server 192.168.90.10
R2-BrA(dhcp-config)#ex
R2-BrA(config)#ip dhcp pool pool-Support
R2-BrA(dhcp-config)#network 192.168.40.0 255.255.255.0
R2-BrA(dhcp-config)#default-router 192.168.40.1
R2-BrA(dhcp-config)#dns-server 192.168.90.10
R2-BrA(dhcp-config)#ex
R2-BrA(config)#
```

- Show IP Route

```
R2-BrA>en
Password:
R2-BrA#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
C       10.0.0.4/30 is directly connected, GigabitEthernet0/1
L       10.0.0.5/32 is directly connected, GigabitEthernet0/1
D       10.0.0.8/30 [90/3072] via 10.0.0.6, 00:14:25, GigabitEthernet0/1
D       10.0.0.12/30 [90/3328] via 10.0.0.1, 00:14:25, GigabitEthernet0/0
                     [90/3328] via 10.0.0.6, 00:14:24, GigabitEthernet0/1
D       10.0.0.16/30 [90/3072] via 10.0.0.1, 00:14:25, GigabitEthernet0/0
D    192.168.10.0/24 [90/28416] via 10.0.0.1, 00:14:25, GigabitEthernet0/0
D    192.168.20.0/24 [90/28416] via 10.0.0.1, 00:14:25, GigabitEthernet0/0
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/2.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/2.30
     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/2.40
L       192.168.40.1/32 is directly connected, GigabitEthernet0/2.40
D    192.168.50.0/24 [90/28416] via 10.0.0.6, 00:14:25, GigabitEthernet0/1
D    192.168.60.0/24 [90/28416] via 10.0.0.6, 00:14:25, GigabitEthernet0/1
D    192.168.70.0/24 [90/28672] via 10.0.0.6, 00:14:24, GigabitEthernet0/1
D    192.168.80.0/24 [90/28672] via 10.0.0.6, 00:14:24, GigabitEthernet0/1
D    192.168.90.0/24 [90/28672] via 10.0.0.1, 00:14:25, GigabitEthernet0/0
D    192.168.99.0/24 [90/28672] via 10.0.0.1, 00:14:25, GigabitEthernet0/0
```

R3-BrB :

- Basic Configuration

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3-BrB
R3-BrB(config)#banner motd "AUTHORIZED ACCESS ONLY - GLN NETWORK"
R3-BrB(config)#enable secret class
R3-BrB(config)#service password-encryption
R3-BrB(config)#line con 0
R3-BrB(config-line)#pass cisco
R3-BrB(config-line)#login
R3-BrB(config-line)#exit
R3-BrB(config)#line vty 0 4
R3-BrB(config-line)#pass cisco
R3-BrB(config-line)#login
R3-BrB(config-line)#exit
R3-BrB(config)#exit
R3-BrB#
%SYS-5-CONFIG_I: Configured from console by console

R3-BrB#clock set 9:00:00 28 sep 2026
R3-BrB#
```

- ROAS and WAN interfaces

```
R3-BrB>en
Password:
R3-BrB#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3-BrB(config)#int g 0/2
R3-BrB(config-if)#no shut

R3-BrB(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

R3-BrB(config-if)#ex
R3-BrB(config)#int g 0/2.50
R3-BrB(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2.50, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.50, changed state to up

R3-BrB(config-subif)#encapsulation dot1q 50
R3-BrB(config-subif)#ip add 192.168.50.1 255.255.255.0
R3-BrB(config-subif)#ex
R3-BrB(config)#int g 0/2.60
R3-BrB(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2.60, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.60, changed state to up

R3-BrB(config-subif)#encapsulation dot1q 60
R3-BrB(config-subif)#ip add 192.168.60.1 255.255.255.0
R3-BrB(config-subif)#ex
R3-BrB(config)#int g 0/1
R3-BrB(config-if)#ip add 10.0.0.6 255.255.255.252
R3-BrB(config-if)#no shut

R3-BrB(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R3-BrB(config-if)#ex
R3-BrB(config)#int g 0/0
R3-BrB(config-if)#ip add 10.0.0.9 255.255.255.252
R3-BrB(config-if)#no shut

R3-BrB(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

R3-BrB(config-if)#ex
R3-BrB(config)#
```

- EIGRP Configurations

```
R3-BrB>en
Password:
R3-BrB#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3-BrB(config)#router eigrp 100
R3-BrB(config-router)#no auto
R3-BrB(config-router)#network 192.168.50.0
R3-BrB(config-router)#network 192.168.60.0
R3-BrB(config-router)#network 10.0.0.4
R3-BrB(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.0.5 (GigabitEthernet0/1) is up: new adjacency

R3-BrB(config-router)#network 10.0.0.8
R3-BrB(config-router)#ex
R3-BrB(config)#
```

- DHCP and DNS Servers

```
R3-BrB>en
Password:
R3-BrB#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3-BrB(config)#ip dhcp excluded-address 192.168.50.0 192.168.50.10
R3-BrB(config)#ip dhcp excluded-address 192.168.60.0 192.168.60.10
R3-BrB(config)#ip dhcp pool pool-Marketing
R3-BrB(dhcp-config)#network 192.168.50.0 255.255.255.0
R3-BrB(dhcp-config)#default-router 192.168.50.1
R3-BrB(dhcp-config)#dns-server 192.168.90.10
R3-BrB(dhcp-config)#ex
R3-BrB(config)#ip dhcp pool pool-Dev
R3-BrB(dhcp-config)#network 192.168.60.0 255.255.255.0
R3-BrB(dhcp-config)#default-router 192.168.50.1
R3-BrB(dhcp-config)#dns-server 192.168.90.10
R3-BrB(dhcp-config)#ex
R3-BrB(config)#
```

- Show IP Route

```
R3-BrB>en
Password:
R3-BrB#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
D       10.0.0.0/30 [90/3072] via 10.0.0.5, 00:14:58, GigabitEthernet0/1
C       10.0.0.4/30 is directly connected, GigabitEthernet0/1
L       10.0.0.6/32 is directly connected, GigabitEthernet0/1
C       10.0.0.8/30 is directly connected, GigabitEthernet0/0
L       10.0.0.9/32 is directly connected, GigabitEthernet0/0
D       10.0.0.12/30 [90/3072] via 10.0.0.10, 00:14:58, GigabitEthernet0/0
D       10.0.0.16/30 [90/3328] via 10.0.0.10, 00:14:58, GigabitEthernet0/0
                     [90/3328] via 10.0.0.5, 00:14:57, GigabitEthernet0/1
D    192.168.10.0/24 [90/28672] via 10.0.0.5, 00:14:57, GigabitEthernet0/1
D    192.168.20.0/24 [90/28672] via 10.0.0.5, 00:14:57, GigabitEthernet0/1
D    192.168.30.0/24 [90/28416] via 10.0.0.5, 00:14:58, GigabitEthernet0/1
D    192.168.40.0/24 [90/28416] via 10.0.0.5, 00:14:58, GigabitEthernet0/1
     192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.50.0/24 is directly connected, GigabitEthernet0/2.50
L       192.168.50.1/32 is directly connected, GigabitEthernet0/2.50
     192.168.60.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.60.0/24 is directly connected, GigabitEthernet0/2.60
L       192.168.60.1/32 is directly connected, GigabitEthernet0/2.60
D    192.168.70.0/24 [90/28416] via 10.0.0.10, 00:14:58, GigabitEthernet0/0
D    192.168.80.0/24 [90/28416] via 10.0.0.10, 00:14:58, GigabitEthernet0/0
D    192.168.90.0/24 [90/28672] via 10.0.0.10, 00:14:58, GigabitEthernet0/0
D    192.168.99.0/24 [90/28672] via 10.0.0.10, 00:14:58, GigabitEthernet0/0

R3-BrB#
```

R4-Whse :

- Basic Configurations

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R4-Whse
R4-Whse(config)#banner motd "AUTHORIZED ACCESS ONLY - GLN NETWORK"
R4-Whse(config)#enable secret class
R4-Whse(config)#service password-encryption
R4-Whse(config)#line con 0
R4-Whse(config-line)#pass cisco
R4-Whse(config-line)#login
R4-Whse(config-line)#exit
R4-Whse(config)#line vty 0 4
R4-Whse(config-line)#pass cisco
R4-Whse(config-line)#login
R4-Whse(config-line)#exit
R4-Whse(config)#exit
R4-Whse#
%SYS-5-CONFIG_I: Configured from console by console

R4-Whse#clock set 9:00:00 28 sep 2026
R4-Whse#
```

- ROAS and WAN interfaces

```
R4-Whse#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4-Whse(config)#int g0/2
R4-Whse(config-if)#no shut

R4-Whse(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
ex
R4-Whse(config)#ex
R4-Whse#
%SYS-5-CONFIG_I: Configured from console by console

R4-Whse#int g0/2.70
          ^
% Invalid input detected at '^' marker.

R4-Whse#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4-Whse(config)#int g0/2.70
R4-Whse(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2.70, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.70, changed state to up

R4-Whse(config-subif)#encapsulation dot1q 70
R4-Whse(config-subif)#ip add 192.168.70.1 255.255.255.0
R4-Whse(config-subif)#ex
R4-Whse(config)#int g0/2.80
R4-Whse(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2.80, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.80, changed state to up

R4-Whse(config-subif)#ip add 192.168.80.1 255.255.255.0

% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL vLAN.

R4-Whse(config-subif)#encapsulation dot1q 80
R4-Whse(config-subif)#ip add 192.168.80.1 255.255.255.0
R4-Whse(config-subif)#ex
R4-Whse(config)#int g0/1
R4-Whse(config-if)#ip add 10.0.0.10 255.255.255.252
R4-Whse(config-if)#no shut

R4-Whse(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
```

- EIGRP Configurations

```
R4-Whse>en
Password:
R4-Whse#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4-Whse(config)#router eigrp 100
R4-Whse(config-router)#no auto
R4-Whse(config-router)#network 192.168.70.0
R4-Whse(config-router)#network 192.168.80.0
R4-Whse(config-router)#network 10.0.0.8
R4-Whse(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.0.9 (GigabitEthernet0/1) is up: new adjacency

R4-Whse(config-router)#network 10.0.0.12
R4-Whse(config-router)#ex
R4-Whse(config)#
```

- DHCP and DNS Servers

```
Enter configuration commands, one per line.  End with CNTL/Z.
R4-Whse(config)#ip dhcp excluded-address 192.168.70.0 192.168.70.10
R4-Whse(config)#ip dhcp excluded-address 192.168.80.0 192.168.80.10
R4-Whse(config)#ip dhco pool pool-Inventory
                       ^
% Invalid input detected at '^' marker.

R4-Whse(config)#ip dhcp pool pool-Inventory
R4-Whse(dhcp-config)#network 192.168.70.0 255.255.255.0
R4-Whse(dhcp-config)#default-router 192.168.70.1
R4-Whse(dhcp-config)#dns-server 192.168.90.10
R4-Whse(dhcp-config)#ex
R4-Whse(config)#ip dhcp pool pool-Shipping
R4-Whse(dhcp-config)#network 192.168.80.0 255.255.255.0
R4-Whse(dhcp-config)#default-router 192.168.80.1
R4-Whse(dhcp-config)#dns-server 192.168.90.10
R4-Whse(dhcp-config)#ex
R4-Whse(config)#
```

- Show IP Route

```
R4-Whse>en
Password:
R4-Whse#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
D       10.0.0.0/30 [90/3328] via 10.0.0.9, 00:15:25, GigabitEthernet0/1
                    [90/3328] via 10.0.0.14, 00:15:25, GigabitEthernet0/0
D       10.0.0.4/30 [90/3072] via 10.0.0.9, 00:15:25, GigabitEthernet0/1
C       10.0.0.8/30 is directly connected, GigabitEthernet0/1
L       10.0.0.10/32 is directly connected, GigabitEthernet0/1
C       10.0.0.12/30 is directly connected, GigabitEthernet0/0
L       10.0.0.13/32 is directly connected, GigabitEthernet0/0
D       10.0.0.16/30 [90/3072] via 10.0.0.14, 00:15:25, GigabitEthernet0/0
D    192.168.10.0/24 [90/28672] via 10.0.0.14, 00:15:25, GigabitEthernet0/0
D    192.168.20.0/24 [90/28672] via 10.0.0.14, 00:15:25, GigabitEthernet0/0
D    192.168.30.0/24 [90/28672] via 10.0.0.9, 00:15:25, GigabitEthernet0/1
D    192.168.40.0/24 [90/28672] via 10.0.0.9, 00:15:25, GigabitEthernet0/1
D    192.168.50.0/24 [90/28416] via 10.0.0.9, 00:15:25, GigabitEthernet0/1
D    192.168.60.0/24 [90/28416] via 10.0.0.9, 00:15:25, GigabitEthernet0/1
     192.168.70.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.70.0/24 is directly connected, GigabitEthernet0/2.70
L       192.168.70.1/32 is directly connected, GigabitEthernet0/2.70
     192.168.80.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.80.0/24 is directly connected, GigabitEthernet0/2.80
L       192.168.80.1/32 is directly connected, GigabitEthernet0/2.80
D    192.168.90.0/24 [90/28416] via 10.0.0.14, 00:15:25, GigabitEthernet0/0
D    192.168.99.0/24 [90/28416] via 10.0.0.14, 00:15:25, GigabitEthernet0/0

R4-Whse#
```

R5-DC :

Basic Configurations :

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R5-DC
R5-DC(config)#banner motd "AUTHORIZED ACCESS ONLY - GLN NETWORK"
R5-DC(config)#enable secret class
R5-DC(config)#service password-encryption
R5-DC(config)#line con 0
R5-DC(config-line)#pass cisco
R5-DC(config-line)#login
R5-DC(config-line)#exit
R5-DC(config)#line vty 0 4
R5-DC(config-line)#pass cisco
R5-DC(config-line)#login
R5-DC(config-line)#exit
R5-DC(config)#exit
R5-DC#
%SYS-5-CONFIG_I: Configured from console by console

R5-DC#clock set 9:00:00 28 sep 2026
R5-DC#
```

ROAS and WAN interfaces

```
R5-DC(config)#int g0/2
R5-DC(config-if)#no shut

R5-DC(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

R5-DC(config-if)#ex
R5-DC(config)#int g0/2.90
R5-DC(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2.90, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.90, changed state to up

R5-DC(config-subif)#encapsulation dot1q 90
R5-DC(config-subif)#ip 192.168.90.1 255.255.255.0
                       ^
% Invalid input detected at '^' marker.

R5-DC(config-subif)#ip add 192.168.90.1 255.255.255.0
R5-DC(config-subif)#ex
R5-DC(config)#int g0/2.99
R5-DC(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2.99, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.99, changed state to up

R5-DC(config-subif)#encapsulation dot1q 99
R5-DC(config-subif)#ip add 192.168.99.1 255.255.255.0
R5-DC(config-subif)#ex
R5-DC(config)#int g0/1
R5-DC(config-if)#ip add 10.0.0.14 255.255.255.252
R5-DC(config-if)#no shut

R5-DC(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R5-DC(config-if)#ex
R5-DC(config)#int g0/0
R5-DC(config-if)#ip add 10.0.0.17 255.255.255.252
R5-DC(config-if)#no shut

R5-DC(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

EIGRP Configurations

```
R5-DC(config)#
R5-DC(config)#router eigrp 100
R5-DC(config-router)#no auto
R5-DC(config-router)#network 192.168.90.0
R5-DC(config-router)#network 192.168.99.0
R5-DC(config-router)#network 10.0.0.12
R5-DC(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.0.18 (GigabitEthernet0/0) is up: new adjacency

%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.0.13 (GigabitEthernet0/1) is up: new adjacency

R5-DC(config-router)#network 10.0.0.16
R5-DC(config-router)#ex
R5-DC(config)#
```

DHCP and DNS Servers

```
R5-DC>en
Password:
R5-DC#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R5-DC(config)#ip dhcp excluded-address 192.168.90.0 192.168.90.10
R5-DC(config)#ip dhcp excluded-address 192.168.99.0 192.168.99.10
R5-DC(config)#op dhcp pool pool-Servers
             ^
% Invalid input detected at '^' marker.

R5-DC(config)#ip dhcp pool pool-Servers
R5-DC(dhcp-config)#network 192.168.90.0 255.255.255.0
R5-DC(dhcp-config)#default-router 192.168.90.1
R5-DC(dhcp-config)#dns-server 192.168.90.10
R5-DC(dhcp-config)#ex
R5-DC(config)#ip dhcp pool pool-Management
R5-DC(dhcp-config)#network 192.168.99.0 255.255.255.0
R5-DC(dhcp-config)#default-router 192.168.99.1
R5-DC(dhcp-config)#dns-server 192.168.90.1
R5-DC(dhcp-config)#ex
R5-DC(config)#
```

Security

```
R5-DC#config ter
Enter configuration commands, one per line.  End with CNTL/Z.
R5-DC(config)#acc
R5-DC(config)#access-list 10 den
R5-DC(config)#access-list 10 deny 192.168.70.0 0.0.0.255
R5-DC(config)#access-list 10 permit any
R5-DC(config)#interface g0/2.90
R5-DC(config-subif)#ip ac
R5-DC(config-subif)#ip access-group 10 out
R5-DC(config-subif)#exit
R5-DC(config)#do write
Building configuration...
[OK]
```

```
R5-DC#
R5-DC#show access-lists
Standard IP access list 10
    10 deny 192.168.70.0 0.0.0.255 (8 match(es))
    20 permit any (11 match(es))

R5-DC#
```

Show IP Route

```
R5-DC#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
D       10.0.0.0/30 [90/3072] via 10.0.0.18, 00:15:51, GigabitEthernet0/0
D       10.0.0.4/30 [90/3328] via 10.0.0.13, 00:15:51, GigabitEthernet0/1
                    [90/3328] via 10.0.0.18, 00:15:51, GigabitEthernet0/0
D       10.0.0.8/30 [90/3072] via 10.0.0.13, 00:15:51, GigabitEthernet0/1
C       10.0.0.12/30 is directly connected, GigabitEthernet0/1
L       10.0.0.14/32 is directly connected, GigabitEthernet0/1
C       10.0.0.16/30 is directly connected, GigabitEthernet0/0
L       10.0.0.17/32 is directly connected, GigabitEthernet0/0
D    192.168.10.0/24 [90/28416] via 10.0.0.18, 00:15:51, GigabitEthernet0/0
D    192.168.20.0/24 [90/28416] via 10.0.0.18, 00:15:51, GigabitEthernet0/0
D    192.168.30.0/24 [90/28672] via 10.0.0.18, 00:15:51, GigabitEthernet0/0
D    192.168.40.0/24 [90/28672] via 10.0.0.18, 00:15:51, GigabitEthernet0/0
D    192.168.50.0/24 [90/28672] via 10.0.0.13, 00:15:51, GigabitEthernet0/1
D    192.168.60.0/24 [90/28672] via 10.0.0.13, 00:15:51, GigabitEthernet0/1
D    192.168.70.0/24 [90/28416] via 10.0.0.13, 00:15:51, GigabitEthernet0/1
D    192.168.80.0/24 [90/28416] via 10.0.0.13, 00:15:51, GigabitEthernet0/1
     192.168.90.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.90.0/24 is directly connected, GigabitEthernet0/2.90
L       192.168.90.1/32 is directly connected, GigabitEthernet0/2.90
     192.168.99.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.99.0/24 is directly connected, GigabitEthernet0/2.99
L       192.168.99.1/32 is directly connected, GigabitEthernet0/2.99

R5-DC#
R5-DC#
```

Switches’ Configurations

SW1-HQ :

The Basic Configurations

```
Switch>
Switch>EN
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW1-HQ
SW1-HQ(config)#banner motd "AUTHORIZED ACCESS ONLY - GLN NETWORK"
SW1-HQ(config)#enable secret class
SW1-HQ(config)#service password-encryption
SW1-HQ(config)#enable secret class
SW1-HQ(config)#line con 0
SW1-HQ(config-line)#pass cisco
SW1-HQ(config-line)#login
SW1-HQ(config-line)#exit
SW1-HQ(config)#line vty 0 4
SW1-HQ(config-line)#login
% Login disabled on line 1, until 'password' is set
% Login disabled on line 2, until 'password' is set
% Login disabled on line 3, until 'password' is set
% Login disabled on line 4, until 'password' is set
% Login disabled on line 5, until 'password' is set
SW1-HQ(config-line)#pass cisco
SW1-HQ(config-line)#login
SW1-HQ(config-line)#exit
SW1-HQ(config)#exit
SW1-HQ#
%SYS-5-CONFIG_I: Configured from console by console

SW1-HQ#clack set 9:00:00 28 sep 2026
           ^
% Invalid input detected at '^' marker.

SW1-HQ#clock set 9:00:00 28 sep 2026
SW1-HQ#
```

Switch’s Vlans

```
SW1-HQ>en
Password:
SW1-HQ#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1-HQ(config)#vlan 10
SW1-HQ(config-vlan)#name Admin
SW1-HQ(config-vlan)#exit
SW1-HQ(config)#vlan 20
SW1-HQ(config-vlan)#name Finance
SW1-HQ(config-vlan)#exit
SW1-HQ(config)#int range f0/1-10
SW1-HQ(config-if-range)#switch mode access
SW1-HQ(config-if-range)#switchport access vlan 10
SW1-HQ(config-if-range)#exit
SW1-HQ(config)#int range f0/11-20
SW1-HQ(config-if-range)#switch mode access
SW1-HQ(config-if-range)#switchport access vlan 20
SW1-HQ(config-if-range)#ex
SW1-HQ(config)#do write
Building configuration...
[OK]
SW1-HQ(config)#int range f0/1
SW1-HQ(config-if-range)#switchport mode trunk

SW1-HQ(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

Show Vlan Brief

```
SW1-HQ#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   Admin                            active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10
20   Finance                          active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1-HQ#
```

SW2-BrA :

The Basic Configurations

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname
% Incomplete command.
Switch(config)#hostname Sw2-BrA
Sw2-BrA(config)#banner motd "AUTHORIZED ACCESS ONLY - GLN NETWORK"
Sw2-BrA(config)#enable secret class
Sw2-BrA(config)#service password-encryption
Sw2-BrA(config)#line con 0
Sw2-BrA(config-line)#pass cisco
Sw2-BrA(config-line)#exit
Sw2-BrA(config)#line vty 0 4
Sw2-BrA(config-line)#pass cisco
Sw2-BrA(config-line)#login
Sw2-BrA(config-line)#ex
% Ambiguous command: "ex"
Sw2-BrA(config-line)#exit
Sw2-BrA(config)#line con 0
Sw2-BrA(config-line)#pass cisco
Sw2-BrA(config-line)#login
Sw2-BrA(config-line)#exit
Sw2-BrA(config)#exit
Sw2-BrA#
%SYS-5-CONFIG_I: Configured from console by console

Sw2-BrA#clock set 9:00:00 28 sep 2026
Sw2-BrA#
```

Switch’s Vlans

```
Sw2-BrA>en
Password:
Password:
Sw2-BrA#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Sw2-BrA(config)#vlan 30
Sw2-BrA(config-vlan)#name Sales
Sw2-BrA(config-vlan)#exit
Sw2-BrA(config)#vlan 40
Sw2-BrA(config-vlan)#name Support
Sw2-BrA(config-vlan)#int f0/1-10
                     ^
% Invalid input detected at '^' marker.

Sw2-BrA(config-vlan)#exit
Sw2-BrA(config)#int f0/1-10
                       ^
% Invalid input detected at '^' marker.

Sw2-BrA(config)#int range f0/1-10
Sw2-BrA(config-if-range)#switch mode access
Sw2-BrA(config-if-range)#switchport access vlav 30
                                               ^
% Invalid input detected at '^' marker.

Sw2-BrA(config-if-range)#switchport access vlan 30
Sw2-BrA(config-if-range)#exit
Sw2-BrA(config)#int range f0/11-20
Sw2-BrA(config-if-range)#switch mode access
Sw2-BrA(config-if-range)#switchport access vlan 40
Sw2-BrA(config-if-range)#ex
Sw2-BrA(config)#int range f0/1
Sw2-BrA(config-if-range)#switchport mode trunk
Sw2-BrA(config-if-range)#ex
Sw2-BrA(config)#do write
Building configuration...
[OK]
Sw2-BrA(config)#
```

Show Vlan Brief

```
SW2-BrA>en
Password:
SW2-BrA#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
30   Sales                            active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10
40   Support                          active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW2-BrA#
```

SW3-BrB :

The Basic Configurations

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Sw3-BrB
Sw3-BrB(config)#banner motd "AUTHORIZED ACCESS ONLY - GLN NETWORK"
Sw3-BrB(config)#enable secret class
Sw3-BrB(config)#service password-encryption
Sw3-BrB(config)#line con 0
Sw3-BrB(config-line)#pass cisco
Sw3-BrB(config-line)#login
Sw3-BrB(config-line)#exit
Sw3-BrB(config)#line vty 0 4
Sw3-BrB(config-line)#pass cisco
Sw3-BrB(config-line)#login
Sw3-BrB(config-line)#exit
Sw3-BrB(config)#exit
Sw3-BrB#
%SYS-5-CONFIG_I: Configured from console by console

Sw3-BrB#clock set 99:00:00 28 sep 2026
                  ^
% Invalid input detected at '^' marker.

Sw3-BrB#clock set 9:00:00 28 sep 2026
Sw3-BrB#
```

Switch’s Vlans

```
AUTHORIZED ACCESS ONLY - GLN NETWORK

User Access Verification

Password:

Sw3-BrB>en
Password:
Sw3-BrB#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Sw3-BrB(config)#vlan 50
Sw3-BrB(config-vlan)#name Marketing
Sw3-BrB(config-vlan)#exit
Sw3-BrB(config)#vlan 60
Sw3-BrB(config-vlan)#name Dev
Sw3-BrB(config-vlan)#exit
Sw3-BrB(config)#int rangr f0/1-10
                         ^
% Invalid input detected at '^' marker.

Sw3-BrB(config)#int range f0/1-10
Sw3-BrB(config-if-range)#switch mode access
Sw3-BrB(config-if-range)#switchport access vlan 50
Sw3-BrB(config-if-range)#exit
Sw3-BrB(config)#
Sw3-BrB(config)#int range f0/11-20
Sw3-BrB(config-if-range)#switch mode access
Sw3-BrB(config-if-range)#switchport access vlan 60
Sw3-BrB(config-if-range)#exit
Sw3-BrB(config)#int range f0/1
Sw3-BrB(config-if-range)#switchport mode trunk
Sw3-BrB(config-if-range)#
```

Show Vlan Brief

```
R3-BrB>en
Password:
R3-BrB#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
50   Marketing                        active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10
60   Dev                              active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
R3-BrB#
```

Sw4-Whse :

The Basic Configurations

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R4-Whse
R4-Whse(config)#banner motd "AUTHORIZED ACCESS ONLY - GLN NETWORK"
R4-Whse(config)#enable secret class
R4-Whse(config)#service password-encryption
R4-Whse(config)#line con 0
R4-Whse(config-line)#pass cisco
R4-Whse(config-line)#login
R4-Whse(config-line)#exit
R4-Whse(config)#line vty 0 4
R4-Whse(config-line)#pass cisco
R4-Whse(config-line)#login
R4-Whse(config-line)#exit
R4-Whse(config)#exit
R4-Whse#
%SYS-5-CONFIG_I: Configured from console by console

R4-Whse#clock set 9:00:00 28 sep 2026
R4-Whse#
```

Switch’s Vlans

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Sw4-Whse
Sw4-Whse(config)#banner motd "AUTHORIZED ACCESS ONLY - GLN NETWORK"
Sw4-Whse(config)#enable secret class
Sw4-Whse(config)#service password-encryption
Sw4-Whse(config)#line con 0
Sw4-Whse(config-line)#pass cisco
Sw4-Whse(config-line)#login
Sw4-Whse(config-line)#exit
Sw4-Whse(config)#line vty 0 4
Sw4-Whse(config-line)#pass cisco
Sw4-Whse(config-line)#login
Sw4-Whse(config-line)#exit
Sw4-Whse(config)#exit
Sw4-Whse#
%SYS-5-CONFIG_I: Configured from console by console

Sw4-Whse#clock set 9:00:00 28 sep 2026
Sw4-Whse#s
```

Show Vlan Brief

```
R4-Whse>en
Password:
R4-Whse#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
70   Inventory                        active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10
80   Shipping                         active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
R4-Whse#
```

Sw5-DC

The Basic Configurations

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R5-DC
R5-DC(config)#banner motd "AUTHORIZED ACCESS ONLY - GLN NETWORK"
R5-DC(config)#enable secret class
R5-DC(config)#service password-encryption
R5-DC(config)#line con 0
R5-DC(config-line)#pass cisco
R5-DC(config-line)#login
R5-DC(config-line)#exit
R5-DC(config)#line vty 0 4
R5-DC(config-line)#pass cisco
R5-DC(config-line)#login
R5-DC(config-line)#exit
R5-DC(config)#exit
R5-DC#
%SYS-5-CONFIG_I: Configured from console by console

R5-DC#clock set 9:00:00 28 sep 2026
R5-DC#
```

Switch’s Vlans

```
AUTHORIZED ACCESS ONLY - GLN NETWORK

User Access Verification

Password:

Sw5-DC>en
Password:
Sw5-DC#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Sw5-DC(config)#vlan 90
Sw5-DC(config-vlan)#name Servers
Sw5-DC(config-vlan)#exit
Sw5-DC(config)#vlan 99
Sw5-DC(config-vlan)#name Management
Sw5-DC(config-vlan)#ex
Sw5-DC(config)#int f0/1-10
                      ^
% Invalid input detected at '^' marker.

Sw5-DC(config)#int range f0/1-10
Sw5-DC(config-if-range)#switch mode access
Sw5-DC(config-if-range)#switchport access vlan 90
Sw5-DC(config-if-range)#ex
Sw5-DC(config)#int range f0/11-20
Sw5-DC(config-if-range)#switch mode access
Sw5-DC(config-if-range)#switchport access vlan 99
Sw5-DC(config-if-range)#ex
Sw5-DC(config)#int range f0/1
Sw5-DC(config-if-range)#switchport mode trunk
Sw5-DC(config-if-range)#
```

Show Vlan Brief

```
R5-DC>en
Password:
R5-DC#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
90   Servers                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10
99   Management                       active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
R5-DC#
```

Successful PING from HQ Admin PC to Warehouse Shipping PC

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::209:7CFF:FE7D:DCE1
   IPv6 Address.....: ::
   IPv4 Address.....: 192.168.10.13
   Subnet Mask.....: 255.255.255.0
   Default Gateway.....: ::
                                 192.168.10.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: ::
   IPv6 Address.....: ::
   IPv4 Address.....: 0.0.0.0
   Subnet Mask.....: 0.0.0.0
   Default Gateway.....: ::
                                 0.0.0.0

C:\>
```

Physical Config Desktop Programming Attributes

Command Prompt

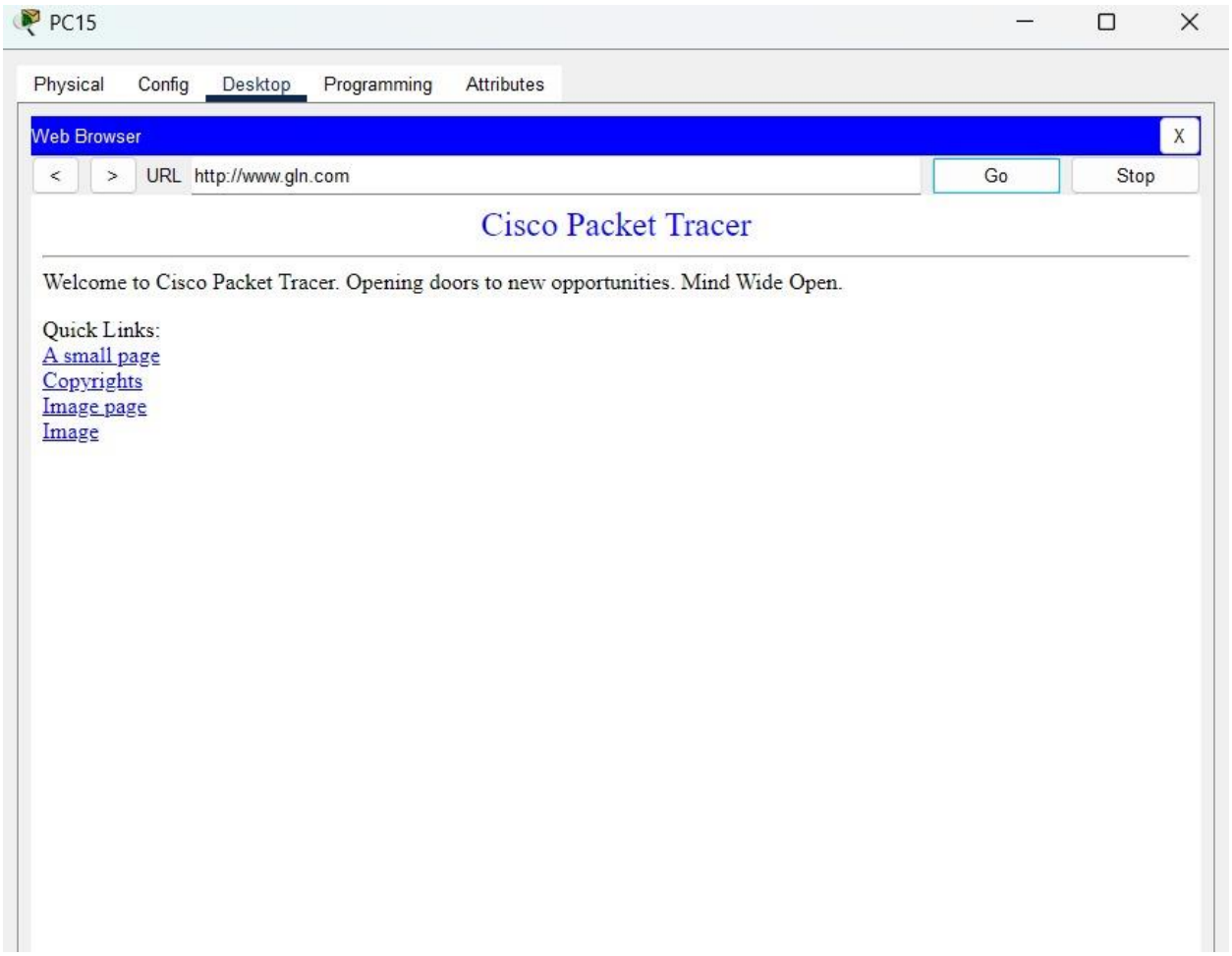
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.90.10

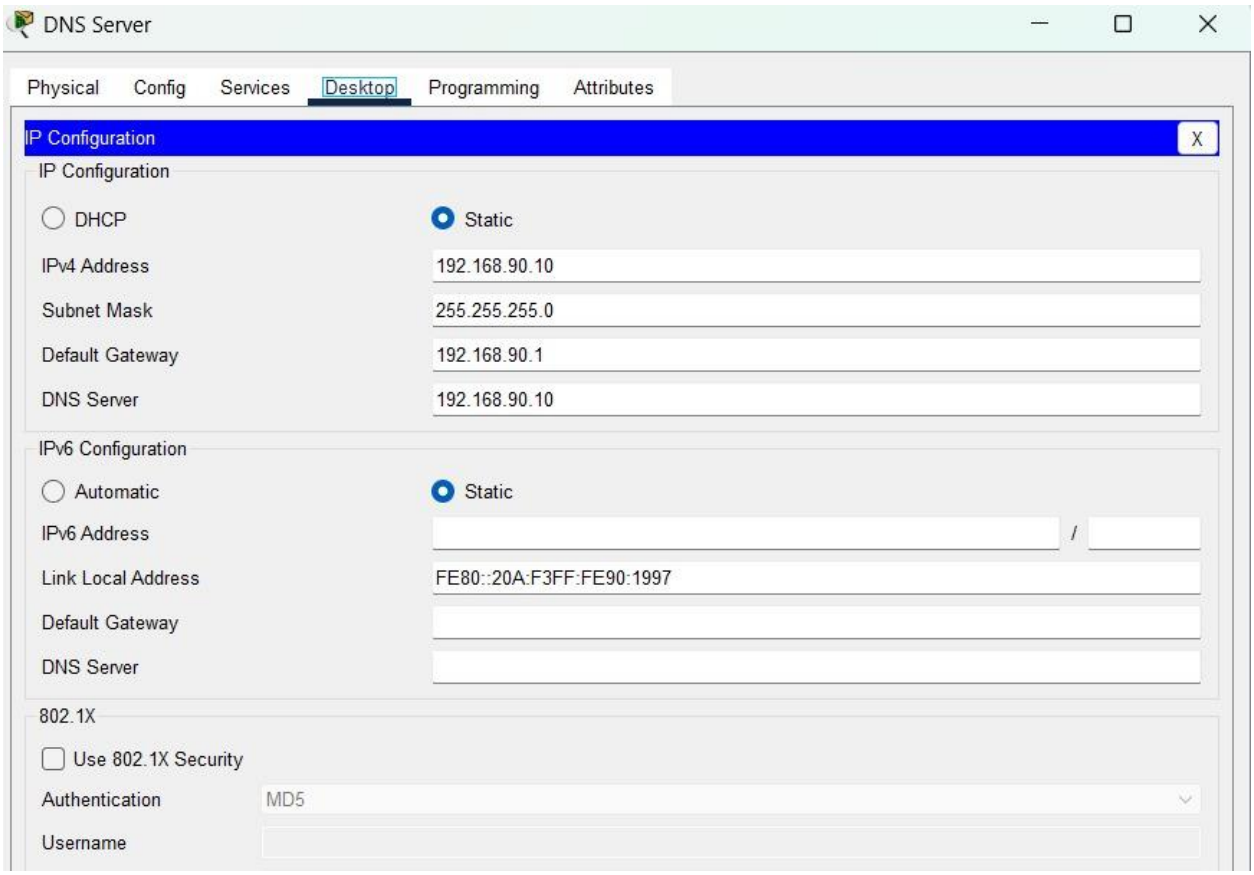
Pinging 192.168.90.10 with 32 bytes of data:

Reply from 10.0.0.14: Destination host unreachable.
Reply from 10.0.0.14: Destination host unreachable.
Reply from 10.0.0.14: Destination host unreachable.
Reply from 10.0.0.14: Destination host unreachable.

Ping statistics for 192.168.90.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```


PC15
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://www.gln.com
Go
Stop
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image


DNS Server
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
192.168.90.10
Subnet Mask
255.255.255.0
Default Gateway
192.168.90.1
DNS Server
192.168.90.10
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::20A:F3FF:FE90:1997
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username